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SUPERSEDING
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MILITARY SPECIFICATION
FOR
RADAR SET AN/APS-80 (V)

This specification has been approved by the
Bureau of Naval Weapons, Department of the Navy

1. SCOPE

1.1 Scope - The equipment covered by this specification shall be an airborne search radar and shall provide means for searching for and detecting surface targets. The equipment can be used as a dual or single radar system.

1.2 Classification - The equipment covered by this specification shall be of one type and shall consist of the following:

Units	Type Designation	Max. Allow. Size (Inches) H	W	L	Max. Allow. Wt. (Lbs.)	Appl. Par.
Antenna	AS-1000/APS-80	46-3/4	45-1/2	32	98.0 (ea)	3.5.1
Amplifier, Electronic Control	AM-2135/APS-80	8-1/8	5-5/8	5	7.8 (ea)	3.5.2
Amplifier Synchro Signal	AM-2719/APS-80	5	10-1/2	25	24.0	3.5.3
Coupler Directional	CU-733/APS-80	2-5/16	2-7/8	7	1.5 (ea)	3.5.4
Compressor, Rotary, Power Driven	HD-358/AP	5-1/2	7-5/8	10-1/4	7.5 (ea)	3.5.5
Receiver-Transmitter, Radar	RT-508/APS-80	20-5/8	16*	16*	69.0 (ea)	3.5.6
Control, Antenna	C-2785/APS-80	5-5/8	5-3/4	7-1/4	6.2 (ea)	3.5.7

*Includes mounting bracket

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Units	Type Designation	Max. Allow. Size (Inches) H	W	L	Max. Allow. Wt. (Lbs.)	Appl. Par.
Control, Radar Set	C-2786/APS-80	9	5-3/4	6	5.0 (ea.)	3.5.8
Modulator-Power Supply	MD-348/APS-80	22-3/4	17-1/2*	17-1/2*	63.0 (ea)	3.5.9
Synchronizer, Electrical	SN-234/APS-80	7-5/8	10-1/8	13-7/8	19.5	3.5.10
Gyroscope, Displacement	CN-549/APS-80 or CN-611/APS-80A	5-7/8	6-1/2	9-1/8	9.0 (ea)	3.5.11
Fan, Tube-axial	HD-359/APS-80	7-5/8	5-5/16	6-11/16	3.0 (ea)	3.5.12
Power Supply	PP-2192/APA-125A	10-3/4	13	19-1/4	40.0 (ea)	3.5.13
Waveguide Switch	SA-512U	4-7/8	4-5/16	6-1/16	5.0 (ea)	3.5.14
Dummy Load Electrical	DA-148/U	3-1/4	2-5/8	6-1/2	1.5 (ea)	3.5.15
Gyroscope, Rate	CN-550/APS-80 or CN-612/APS-80A	4-3/16	3-1/4	5-7/16	2.5 (ea)	3.5.16
≠Interconnecting Box	J-1192/APS-80	4	10	12-1/4	8.0 (ea)	3.5.17
≠Interconnecting Box	J-1035/APS-80	25	10	3	15.0	3.5.18
Transformer Variable Power	TF-284/APS-80	7-3/8	9-1/8	12-1/8	20.0 (ea)	3.5.19
Mounting	MT-2134/APS-80	12	17-1/2	17-1/2	8.0 (ea)	3.5.6.4
Mounting	MT-2135/APS-80	5-1/2	20 diam.		6.5 (ea)	3.5.9.4
Mounting	MT-2136/APS-80	3	11-1/4	15-3/4	2.5	3.5.10.2
Mounting	MT-1824/APA-125	3-1/2	13	20	2.5 (ea)	3.5.13.2

≠ Note: Dual Radar Set AN/APS-80 uses Amplifier Synchro Signal AM-2719/APS-80 and Interconnecting Box J-1192/APS-80. For use as a single AN/APS-80 Radar Set delete AM-2719/APS-80 and replace Interconnecting Box J-1192/APS-80 with J-1035/APS-80.

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1.3 Accessories - The Radar Set AN/APS-80 shall operate with the following accessories and associated equipment which shall not be supplied as a part of this equipment:

Radar Recognition Set	AN/APX-7
Computer, Group, Navigational	AN/ASA-13
Radio Receiving Set	AN/ARR-58
Indicator Group	AN/APA-125A
Test Set, Radar	AN/UPM-32
Data Display Group	AN/ASA-16

2. APPLICABLE DOCUMENTS

2.1 General - The following documents of the issue in effect on the date of invitation for bids, form a part of this specification.

SPECIFICATIONS

Military

MIL-C-172	Cases; Bases, Mounting, and Mounts, Vibration (For use with Electronic Equipment in Aircraft)
MIL-E-4682	Electron Tubes, Choice and Application of
MIL-W-5088	Wiring, Aircraft, Installation of
MIL-E-5400	Electronic Equipment, Airborne, General Specification for
MIL-T-5422	Testing; Environmental, Aircraft Electronic Equipment
MIL-I-6181	Interference Control Requirements, Aircraft Equipment
MIL-C-6781	Control Panel; Aircraft Equipment, Rack or Console Mounted
MIL-P-7788	Plate, Plastic, Lighting
MIL-P-17555	Preparation for Delivery of Electronic, Electrical, and Associated Repair Parts and Electro-mechanical Equipment

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MIL-T-18303 Test Procedures, Preproduction and Inspection, for Aircraft Electronic Equipment; Format for

MIL-N-18307 Nomenclature and Nameplates for Aeronautical Electronic and Associated Equipment

MIL-S-20708 60 and 400 Cycle Synchros, General Specification

SAR-300 Material Changes and Material Bulletins, Preparation of

STANDARDS

Military

MIL-STD-704 Electric Power, Aircraft, Characteristics and Utilization of

MS 25245 Connections, Input Power Airborne Electronic Equipment; 28-Volt DC, 115-Volt Single-Phase AC, or 115/200-Volt 3-Phase AC, or Combinations

MS 28053 Meter, Time Totalizing, 400-Cycle

2.2 Availability of Documents

(1) When requesting specifications, standards, drawings, and publications, refer to both title and number. Copies of this specification and applicable specifications required by contractors in connection with specific procurement functions may be obtained upon application to the Commanding Officer, Naval Supply Depot, Code CDS, 5801 Tabor Avenue, Philadelphia 20, Pennsylvania.

3. REQUIREMENTS

3.1 Parts and Materials - In the selection of parts and materials, fulfillment of major design objectives shall be the prime consideration. In so doing, the following shall govern:

(1) Parts and materials requirements shall conform to Specification MIL-E-5400.

(2) Nonrepairable subassemblies, as outlined in specification MIL-E-5400, shall be used when practicable. The general size of the subassembly, and the amount of circuitry to be included therein shall be approved by the procuring activity. Nonrepairable subassemblies must be reliable. A mean-time-to failure of more than 5000 hours should be the goal of each.

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(3) When previously produced models of this equipment did not use nonrepairable subassemblies, the design shall not be changed to employ nonrepairable assemblies without the approval of the procuring activity.

3.1.1 Nonstandard Parts and Material Approval - Approval for the use of nonstandard parts and materials shall be obtained as outlined in Specification MIL-E-5400.

3.1.2 Electron Devices - Electron tubes, transistors and diodes shall be chosen and applied, and the complements reported as outlined in Specification MIL-E-4682. The complement report must be submitted to the procuring activity for review and approval prior to preproduction testing.

3.2 Design and Construction

3.2.1 General - This equipment shall fulfill all applicable requirements of Specification MIL-E-5400 for design, construction and workmanship unless otherwise specified herein.

3.2.2 Service Conditions - The equipment shall operate satisfactorily under any of the environmental and mechanical service conditions or reasonable combination of these conditions as specified in Specification MIL-E-5400 for Class 1 equipment.

3.2.2.1 Sand and Dust - The equipment shall be as dustproof as practicable consistent with cooling requirements.

3.2.2.2 Altitude - The complete equipment shall operate satisfactorily at barometric pressures ranging from 30 in. Hg down to 8.9 in. Hg, (sea level to 30,000 ft. altitude). Transporting the equipment in a non-operating condition at barometric pressures down to 3.4 in. Hg (50,000 ft. altitude) shall cause no permanent damage to the equipment and it shall be possible to resume normal operation when the equipment is returned to pressures equivalent to 30,000 feet or lower altitudes.

3.2.3 Standard Conditions - The following conditions shall be used to establish normal performance requirements and for making laboratory bench tests.

Temperature	Room Ambient (30°C ±10°C)
Altitude	Normal ground
Vibration	None
Humidity	Room ambient up to 90% relative humidity
Input power voltage	3-Phase 4-wire system, 115 ±1.0 V AC line to neutral and 27.5 ±0.5 V DC

3.2.4 Total Weight - The total weight of the equipment, excluding cables, shall be a minimum consistent with good design and shall not exceed 796 pounds for the dual system, and 380 pounds for a single system.

3.2.5 Operational Stability - The equipment shall operate with optimum performance for 24 hours, continuously or intermittently, without the necessity for readjustment of any controls which are inaccessible during flight.

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3.2.6 Operating Life

3.2.6.1 Reliable Operating Life - The equipment shall have a reliable operating life of at least 300 hours without removal for bench servicing. Parts requiring servicing or replacement at the end of this interval to renew this service life shall be specified by the manufacturer.

3.2.6.2 Total Operating Life - The equipment shall have a minimum total operating life of 2000 hours with reasonable servicing and replacement of parts. Parts requiring replacement within this interval and the life of such parts shall be specified by the manufacturer.

3.2.6.3 Maintenance Time Totalizing Meter - An unilluminated 400-cycle time totalizing meter, MX 28053-2 shall be included in the Transformer, Variable Power TF-284/APS-80 for totalizing transmitter operating time.

3.2.7 Interconnection Cabling - This equipment shall be capable of satisfactory operation using external wiring in accordance with the applicable requirements of Specification MIL-W-5088. The external wiring shall be unshielded, except that a minimum number of the individual wires may be shielded when demonstrated as necessary to meet interference control requirements and provided the assembly of the cable to its plugs may be easily accomplished.

3.2.8 Cables and Connectors - The equipment shall provide for the use of cables and connectors in accordance with Specification MIL-E-5400. However, cables and that portion of the connectors which are to be attached to the cables shall not be supplied as part of the equipment.

3.2.9 Control Panels - All rack or console mounted control panels shall conform to the applicable requirements of Specification MIL-C-6781. The configuration of all control panels must be approved by the procuring activity prior to preproduction testing.

*3.2.10 Primary Input Power Requirements - The equipment shall meet all applicable requirements of MIL-STD-704 and shall give specified performance from the following power sources with characteristics as defined in MIL-STD-704 having limits as specified herein. The power required shall not exceed the specified amounts. (See 6.5)

Dual Radar System

(1) AC Power (Three-Phase) 115/200V, Category "B" 5750 VA

(2) DC Power 28V Category "B" 12 Amps

Single Radar System

(1) AC Power (Three-Phase) 115/200V, Category "B" 2750 VA

(2) DC Power 28V Category "B" 6 Amps.

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3.2.11 Input Power Connection - Input power connections shall be in accordance with MS 25245.

3.2.12 Overload Protection - All parts and circuits of the equipment which are likely to carry an overload due to any failures or poor adjustments shall have suitable protective devices, either within or outside the equipment, or shall be proportioned to withstand such overload without permanent damage to the equipment. The use of fuses or other protective devices within the equipment shall be held to a minimum.

3.2.12.1 Undervoltage Protection - The equipment shall not be damaged by voltages below the minimum specified herein and shall automatically resume normal operation when the voltage returns within limits.

3.2.13 Standby Provisions - The equipment shall provide for a standby position wherein rated voltage shall be applied to all tube filaments. In addition, all voltages except the transmitter high voltage shall be normal after approximately a 1-minute time delay.

3.2.14 Fungus Resistant Treatment - Equipments shall be fungus proofed by selection of parts and materials that are non-nutrient for fungus, or the parts and materials shall be treated prior to their use in the equipment so that overall spraying of the equipment is not necessary.

3.2.15 Nomenclature and Nameplates - Nomenclature assignments and nameplate approval for equipment identification shall be in accordance with Specification MIL-N-18307.

3.2.16 Interchangeability - The equipment shall meet the interchangeability requirements as defined in Specification MIL-E-5400.

3.2.17 Maintenance Provisions and Field Testing - Provisions for maintenance shall be as specified in Specification MIL-E-5400. Specific test points and test facilities shall be provided to the greatest extent practicable for ease of field testing and maintenance.

3.2.17.1 Additional Adjustments - Servicing and maintenance adjustments in excess of those now specified shall not be incorporated unless approved by the procuring activity.

3.2.18 Radio Interference Control - The generation of radio interference by the equipment and the vulnerability of the equipment to radio interference shall be controlled within the limits of Specification MIL-I-6181.

3.3 Performance

3.3.1 Warm-Up Time - The warm-up time for the equipment shall not exceed 3 minutes under standard conditions and 8 minutes at extreme service conditions.

3.3.2 Frequency Characteristics - The equipment shall be capable of operating over a frequency range of 8500 to 9600 mcs and the transmitter shall be manually tunable over this range.

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3.3.3 Modulation Characteristics - Each transmitter shall be pulse modulated at a line locked pulse repetition rate of 380 to 420 cps by a pulse width of 2.4 ± 0.12 usec.

3.3.4 Transmitter Power Output - The equipment shall be capable of delivering not less than 143 kw peak measured at each directional coupler.

3.3.5 System Noise Figure - Each overall system noise figure shall not exceed 8.3 db when measured at 9375 megacycles and 10.5 db when measured at frequencies of 8500, 9000 and 9600 mc.

3.3.6 Receiver Sensitivities - The minimum detectable signal shall be at least -107 dbm for each receiver.

3.3.7 Anti-Jam Provisions - The most effective anti-jam design features that can be obtained compatible with other performance requirements shall be incorporated. The contractor shall inform the procuring activity and receive approval of the anti-jam features he proposes to incorporate in the equipment. The contractor shall prescribe the test procedures, as required in 4.6 to demonstrate compliance with this paragraph. The following anti-jam provisions shall be incorporated in the receiving system.

(1) Sensitivity Time Control

(2) IAGC Controlled IF Strip

3.3.8 Displays - Indicator Group AN/APA-125A and AN/ASA-16 which are not supplied as a part of the dual Radar Set AN/APS-80 shall be the indicators for this equipment. This equipment shall operate satisfactorily with both indicators.

3.3.9 IFF Provisions

(1) Display - The display of IFF Video as well as mode of challenge shall be selected by the operation of one of the three IFF selector switches on the panel of the IP-414A/APA-125 Indicator.

(2) Synchronization - Synchronization of the necessary AN/APX-7 IFF equipment shall be accomplished by the SN-234/APS-80 Synchronizer.

3.3.10 System Pressure - Units in each AN/APS-80 Radar Set requiring pressurization shall be interconnected to Compressor, Rotary, Power Driven MD-358/AP. These units shall not be damaged by a pressure of 30 pounds per square inch gage. The units in each pressurized system shall be sealed sufficiently so as not to lose more than 2 PSIG in 4 hours when pressurized to 20 PSIG. A pressure relief valve set to relieve at 24 ± 2 PSIA shall be provided on each pressurizing pump.

3.3.11 Dual Antenna Operation - The antennas shall be controlled by the Antenna Control C-2785/APS-80.

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3.3.12 Single Radar System Operation - By setting either the forward or aft radar system to standby or off, the remaining system shall exhibit 210° sector scan operating characteristics of a single AN/APS-80 Radar Set.

3.4 Workmanship - Workmanship, fabrication, and manufacturing processes shall conform to Specification MIL-E-5400.

3.5 Detail Requirements -

3.5.1 Antenna AS-1000A/APS-80 - These antennas shall meet the following requirements:

3.5.1.1 Function - This antenna shall focus and direct the energy furnished by the radar transmitter. The antenna shall be capable of continuous rotation --- and when selected, of sector scanning; shall provide an output required for North stabilization of indicator presentation as well as an output for relative bearing presentation when supplied with proper excitation signal voltages. Each antenna shall be tilt-stabilized.

3.5.1.1.1 Functional Control - Functional control of the various modes of antenna operation within limits specified herein shall be at the discretion of the operator when using the Control, Antenna C-2785/APS-80.

3.5.1.2 Form Factor and Mounting - The overall dimensions shall not exceed 45-1/2 inches in height, 46-3/4 inches in width and 32 inches overall depth. All mounting surfaces shall be in one plane which shall be perpendicular to the vertical axis of the antenna. Mounting isolators shall be provided and shall be located on 17.375 by 19.375 inch centers.

3.5.1.3 Power Capacity - The RF systems shall be capable of handling 275 kw peaks at sea level atmospheric pressure.

3.5.1.4 Contents - In addition to the radiating system and azimuth drive motor with it's associated control circuits, each antenna shall include the azimuth differential synchro, relative bearing synchro, resolver synchro and tilt servo unit.

3.5.1.5 Pressurization - The RF transmission line systems shall be a part of the pressurized system. Pressurization shall be effective over the reliable operating range of the equipment.

3.5.1.6 Attenuation - The one-way loss for transmitted power through the complete waveguide and rotary joints shall not exceed 0.8 db at any frequency in the 8500-9600 mc range.

3.5.1.7 Voltage Standing Wave Ratio - The VSWR of the antenna microwave components shall not exceed 1.50 when measured at frequencies of 8500, 8720, 8940, 9160, 9375, and 9600 mc. Selection of RF components to meet these requirements shall not be necessary.

3.5.1.8 Reflector - The reflector shall be a nominal 45 by 26 inches in size, having the curvature of a paraboloidal surface.

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3.5.1.8.1 Polarization - The radiated electric field shall be horizontally polarized when the antenna is mounted normally in an erect attitude.

3.5.1.8.2 Horizontal Pattern - The 3 db beamwidth measured at any depression or elevation angle within the range specified for the vertical pattern shall be not more than 2.4° when measured at frequencies of 9000, 9375, and 9600 megacycles and not more than 2.5° when measured at a frequency of 8500 megacycles. Directions of maximum power shall occur at the same azimuth position (within 0.4°) at any depression or elevation angle within the limits specified. Any side lobes shall be down at least 22 db from maximum power when measured at a frequency of 9375 megacycles and shall be down at least 18 db from maximum power when measured at frequencies of 8500, 9000, and 9600 megacycles.

3.5.1.8.3 Vertical Radiation Pattern -

(1) Pencil Beam - The 3 db beamwidth measured at any tilt angle shall be not more than 3.6° when measured at frequencies of 9375 mc and 9600 mc and shall not be more than 3.8° when measured at frequencies of 8500 mc and 9000 mc. The maximum vector shall not vary by more than 0.4° from the mechanical axis of the antenna. Side lobes in the principal plane shall be at least 22 db below the maximum power at any angle of tilt when measured at a frequency of 9375 mc and shall be at least 18 db below the maximum power at any angle of tilt when measured at frequencies of 8500 mc, 9000 mc and 9600 mc.

3.5.1.8.4 Spoiler Plate - A spoiler plate designed for a cosecant squared pattern shall be provided. The plate shall be normally stowed on the back of the antenna and shall be easily attached to and detached from the parabolic reflector.

3.5.1.8.5 Tilt Stabilization - The pre-set reflector tilt shall be automatically maintained within ±0.5° for 0° tilt. Maximum permissible error shall not be greater than 1.0° for any tilt angle up to 30 degrees. Any resultant up to ±60° shall not cause damage to the antenna or stabilization equipment, and the equipment shall operate normally when the resultant returns to within the tilt limits. In addition, the stabilization system shall permit removing the gyroscope information from the circuit by operation of the "STAB-OUT" switch on Control, Antenna C-2785/APS-80. Tilt of the reflector shall be controlled by, and correspond to, the setting of the "TILT" control on Control, Antenna C-2785/APS-80. Overall beam tilt angle limits including normally pre-set values plus stabilization shall be at least 30 degrees depression to 30 degrees elevation. It shall also be possible to lock the reflector mechanically at 0° fixed beam tilt angle when the equipment is not operating. For maintenance checks, it shall be possible to lock the reflector mechanically at +10°, +20°, +30°, -10°, -20°, and -30° tilt angles.

3.5.1.8.6 Continuous Rotation Rate - In response to operating signals, the reflectors shall rotate at a constant rate of 6 ± 1 rpm.

3.5.1.8.7 Sector Width - In response to operating control signals the antenna beam shall scan the 45° sector width to within ±2°. The sector width can be adjusted up to 240° upon installation. It shall be possible to stop the antenna at any azimuth angle within ±1/2°.

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3.5.1.8.8 Sector Scan Rate - In response to operating controls, the reflector shall scan the 45° sector at a rate of 36° per second.

3.5.1.8.9 Sector Location - In response to operating control signals, the center of the sector shall be adjustable through 210°. In addition the reflector shall be capable of 360° scan operation and shall be further capable of full scan adjustable mechanically from 30° to 230°.

3.5.1.8.10 Searchlighting Provision - The antenna reflector shall be capable of being searchlighted; i.e., "stopped", at any position in response to operator control.

3.5.1.8.11 Manual Tilt Control Limits - Beam tilt of the reflector shall cover at least the range of +10 +1° elevation to -20 ±1° depression in response to manual operation of the "TILT" control. The tilt servo system shall cause the beam to achieve a pre-determined tilt position within ±1/4° of the setting of the "TILT" control when approached from either direction.

3.5.1.8.12 Feed - The radio frequency energy shall be delivered to and received from the reflector by means of a horn feed of such design that illumination of the reflector will result in the radiation characteristics specified herein.

3.5.1.8.13 Antenna Gain - The gain of the antenna measured at the first waveguide flange in back of the reflector shall be at least 35 db for pencil beam and 30 db for Csc^2 beam when measured at 9375 megacycles, and shall be at least 34 db for pencil beam when measured at 8500, 9000 and 9600 megacycles.

3.5.1.8.14 Index Marks - The rotary base shall contain azimuth index marks at 45° intervals.

3.5.1.9 Azimuth Differential Synchro - A synchro shall be provided, geared such that its rotor rotates through an angle equal to the angular rotation of the reflector in the same direction. Backlash between the synchro rotor and the reflector shall not exceed 0.3°. This synchro shall be the electrical and mechanical equivalent of commercial type AY-2315-3 or better.

3.5.1.10 Relative Bearing Azimuth Synchro - A suitable synchro shall be provided for relative bearing stabilization of the video display on the radar indicator. Backlash shall not exceed 0.3°. The required synchro shall be the electrical and mechanical equivalent of commercial type AY-201-1 or better.

3.5.1.11 Tilt Stabilization Resolver Synchro - A synchro geared so that its rotor rotates in the same direction as the reflector and through an angle equal to the angular rotation of the reflector shall be provided.

3.5.1.12 Leveling - One or more surfaces, capable of providing areas on which can be placed a spirit level, to establish a plane during installation perpendicular to the axis of azimuth rotation of the reflectory, shall be included on the antenna. These areas shall be labeled "REF LEVEL" and shall represent a plane perpendicular to the axis of azimuth rotation within 0.1°.

3.5.1.13 IFF Antenna -

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3.5.1.13.1 Function - The antenna shall radiate and collect IFF information when operating with compatible equipment such as Recognition Set, Radar, AN/APX-7.

3.5.1.13.2 Power Handling Capability - The RF components shall be capable of handling a peak power of 4 kw.

3.5.1.13.3 Radiation Pattern Characteristics - With the IFF antenna properly located in respect to the reflector of Antenna AS-1000/APS-80, it shall exhibit the following pattern characteristics at any frequency from 1010 to 1030 and 1090 to 1110 mc.

3.5.1.13.3.1 Polarization - The radiated electric field shall be vertically polarized.

3.5.1.13.3.2 Azimuth Plane Beamwidth - The beamwidth at the half power points shall be 23° maximum.

3.5.1.13.3.3 Elevation Plane Beamwidth - The beamwidth at the half power points shall be 45° maximum.

3.5.1.13.4 Antenna Gain - The gain of the antenna shall be at least 12 db over the frequency range of 1010 to 1030 and 1090 to 1110 mc.

3.5.1.13.5 Voltage Standing Wave Ratio - With the IFF antenna assembled on Antenna AS-1000/APS-80 as an operating system, the VSWR measured on the transmitter side of IFF channel of the azimuth rotary joint shall be as follows:

3.5.1.13.5.1 The VSWR shall not exceed 1.70:1 in the frequency range of 1010 to 1030 mc and 2.00:1 in the frequency range of 1090 to 1110 mc.

3.5.1.13.6 IFF and Radar Beam Alignment - In the horizontal or vertical plane, the maximum deviation between the axes of the radar main lobe and the IFF main lobe shall not exceed 3 degrees.

3.5.1.13.7 Interference Between Radar and IFF Equipment - Rejection of signals between the radar and IFF antenna shall be such that optimum use of either system shall not be impaired.

3.5.1.13.8 IFF RF system shall not be pressurized.

3.5.2 Amplifier, Electronic Control AM-2135/APS-80 - Each electronic control amplifier shall meet the following requirements:

3.5.2.1 Function - This unit shall be capable of supplying the control phase driving voltage for either the antenna azimuth drive motor or the antenna tilt drive motor.

3.5.2.2 Dimensions - The overall dimensions shall not exceed 8-1/8 inches in height, 5-5/8 inches in width, and 5 inches in depth.

3.5.2.3 Weight - The weight shall not exceed 7.8 pounds.

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3.5.2.4 Mounting Provisions - This unit shall be mounted on the AS-1000/APS-80.

3.5.2.5 Controls - This servo amplifier shall have externally accessible screwdriver slot type adjustments necessary to insure satisfactory operation of the servo amplifier.

3.5.3 Amplifier, Synchro Signal AM-2719/APS-80 - The synchro signal amplifier shall meet the following requirements:

3.5.3.1 Function - The synchro signal amplifier shall provide necessary wiring and connectors for interconnecting components of Dual Radar Set AN/APS-80 and applicable components of the accessory equipments listed in paragraph 1.3 such that the dual antenna system shall have operating characteristics identical to those of a single AN/APS-80 radar system with the antenna rotating continuously.

3.5.3.2 Form Factor and Mounting - The overall dimensions shall not exceed 5 inches in height, 10-1/2 inches in width and 25 inches in depth. Mounting shall be on a flat surface and be on 5 by 16 inch centers.

3.5.3.3 Weight - The weight shall not exceed 24 pounds.

3.5.3.4 Synchro Drive Assembly - The synchro drive assembly shall perform the relative bearing synchronization of two AS-1000/APS-80 antennas. The synchro drive servo motor shall be driven by the relative bearing error signal from Antenna Control, C-2785/APS-80, and in turn drive the following:

(1) One relative bearing transmitting synchro.

(2) Two relative bearing control transformer synchros.

(3) One cam assembly and associated switches.

Provisions for adjusting synchros and cam actuated switches shall be provided.

3.5.3.4.1 Relative Bearing Transmitter Synchro - This synchro shall be excited from the 400 cycle supply and complete the servo loop between Antenna Control C-2785/APS-80 and Synchro Signal Amplifier AM-2719/APS-80. The required synchro shall be the electrical and mechanical equivalent of commercial type AY-201-1 or better.

3.5.3.4.2 Relative Bearing Control Transformer Synchros - Each of these synchros shall be excited from an azimuth position synchro, B808, of Antenna AS-1000/APS-80 and provide an output sufficient to drive the azimuth Electronic Control Amplifier AM-2135/APS-80, of the same antenna. Both synchros shall be electrically and mechanically equivalent of commercial type AY-201-3 or better.

3.5.3.4.3 Scan Control Switches - Relays used for dual antenna circuit switching shall be operated by cam actuated switches. The configuration of the cam shall be such that the switches will be open (non-conductive) while the forward antenna is in the forward relative bearing sector, 270° to 90°, and closed while in the aft relative bearing sector.

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3.5.3.4.4 Heading Marker Switch - This switch shall close a two-wire circuit to the AN/ASA-16 when the forward antenna crosses the relative bearing of aircraft dead-ahead.

3.5.3.5 Video Relay - This relay shall furnish video to the indicators from the radar receivers. The scan control switches shall operate this relay.

3.5.3.6 Relative-True Relays - Relative heading or true heading sweep voltages for the indicators shall be available through these relays at the discretion of each indicator operator.

3.5.3.7 External Relay - Provisions for switching an external relay shall be provided. This relay will switch the IFF equipment to the aft antenna when selected by the scan control switches described in paragraph 3.5.3.4.3.

3.5.3.8 Cursor 2 Relay - This relay shall switch control of the cursor 2 control transformer in Indicator IP-414A/APA-125 from the cursor 2 synchro transmitter to the aircraft compass. The IP-414A/APA-125 operator shall control the operation of this relay with the "HDG MKR" switch.

3.5.4 Coupler, Directional CU-733/APS-80 - Directional couplers shall be provided to meet the following requirements:

3.5.4.1 Function - The directional coupler shall provide for sampling or inserting small amounts of radio frequency energy in the waveguide between the receiver-transmitter and the antenna.

3.5.4.2 Dimensions - The outline dimensions shall be approximately: height 2-5/6 inches, width 2-7/8 inches and depth 7 inches.

3.5.4.3 Weight - The weight shall not exceed 1.5 pounds.

3.5.4.4 Mounting Provisions - The directional coupler shall be supported by its end connections.

3.5.4.5 Pressurization - The directional coupler shall form part of the pressurized RF transmission line system and shall be tested to 30 pounds per square inch gauge.

3.5.4.6 Attenuation - The one-way insertion loss for transmitted power due to the directional coupler shall not exceed 0.2 db at any frequency in the 8500-9600 mc range.

3.5.4.7 Voltage Standing Wave Ratio - The voltage standing wave ratio shall not exceed 1.15 in the main waveguide.

3.5.4.8 Receptacles - This directional coupler shall have the receptacles and labels as indicated in the following paragraphs:

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3.5.4.8.1 "TEST CONNECTION" - This receptacle shall mate with AN type UG-21/U plug and shall provide ratio frequency power attenuation of 30 db ±1.0 db over the frequency range of 8500-9600 mc down from the power transmitted in the main waveguide. The directivity shall be at least 15 db, where directivity is defined as the ratio of power levels expressed in decibels coupled out by the directional coupler from a line which is matched beyond the coupler when the coupler is inserted first in such a direction as to couple to the transmitted power and then in such a direction as to couple out the reflected power. The actual value of attenuation at 8500, 9000, and 9600 mcs at the test connection shall be prominently labeled near this connection to ±0.1 db.

3.5.4.8.2 "TRANS" - This receptacle shall connect to the transmitter-receiver, shall be a plain flange, and shall be capable of mating with a 1-1/4 inch by 5/8 inch choke flange.

3.5.4.8.3 "ANT" - This receptacle shall connect to the antenna, shall be a plain flange, and shall be capable of mating with a 1-1/4 inch by 5/8 inch choke flange.

3.5.5 Compressor, Rotary, Power Driven HD-358/AP - Each pressurizing unit shall be furnished to meet the following requirements:

3.5.5.1 Function - This unit shall be used to maintain pressure in the pressurized system at 20 ±2 psia up to an altitude of 30,000 feet.

3.5.5.2 Dimensions - The overall dimensions shall not exceed 5-1/2 inches height by 7-5/8 inches width by 10-1/4 inches depth.

3.5.5.3 Weight - Weight shall not exceed 7.5 pounds.

3.5.5.4 Mounting Provisions - This unit may be rigidly mounted on the Antenna AS-1000/APS-80.

3.5.5.5 Operation - This pump shall dehydrate the pressurizing air to a relative humidity up to 40%. When the relative humidity of the ambient air remains above 90%, then the drying agent shall not require reactivation or replacement in less than 15 hours. The drying agent shall be selected such that no small particles can pass into the pressurized system through the filters provided. Relative humidity conditions shall be referred to 25°C dry bulb temperature.

3.5.6 Receiver-Transmitter, Radar RT-508/APS-80 - Each receiver-transmitter shall meet the following requirements:

3.5.6.1 Function - The receiver-transmitter shall generate pulsed X-band RF energy and shall detect echoes received from these pulses.

3.5.6.2 Dimensions - The overall dimensions, including the mounting flanges, shall not exceed 20-5/8 inches height by 16 inches width by 16 inches depth.

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3.5.6.3 Weight - The weight shall not exceed 69 pounds, exclusive of mounting.

3.5.6.4 Mounting - The receiver-transmitter shall be mounted on MT-2134/APS-80, a vibration isolator type shock mounting conforming to the applicable requirements of Specification MIL-C-172.

3.5.6.5 Type of Construction - The receiver-transmitter shall be of cylindrical construction.

3.5.6.6 Pressurization - This unit shall form a part of the pressurized system and shall be capable of withstanding a continuous internal pressure of 30 psi, gauge.

3.5.6.7 Frequency - The search transmitting and receiving frequencies of this equipment shall be manually tunable over the range of 8500-9600 mc. The equipment shall operate on a frequency within this range, selected at random, when delivered.

3.5.6.8 RF Components - The RF components shall consist of Duplexer, Balanced Mixer, Local Oscillator Assembly, and interconnecting waveguide as required.

3.5.6.8.1 Duplexer - The duplexer shall be of short-slot hybrid design containing one dual TR tube.

3.5.6.8.1.1 Standing Wave Ratio - The Voltage Standing Wave Ratio of the duplexer when transmitting shall not exceed 1.25:1. The VSWR of the duplexer in the receiving condition shall not exceed 1.4:1. Both VSWR readings shall be taken at 9460 mc/s.

3.5.6.8.2 Balanced Mixer - The balanced mixer shall contain mounting provision for two 1N23WE crystals, with radar IF outputs in phase. It shall also contain one 1N23WE AFC crystal. A flange shall be provided for mounting the local oscillator tube. The mixer assembly shall include an attenuator for adjusting crystal current.

3.5.6.8.3 Local Oscillator - The local oscillator shall be Varian Type 203-B reflex klystron or approved equivalent.

3.5.6.8.3.1 Local Oscillator Tuning - Means shall be provided for fast and accurate tuning of the radar local oscillator tube cavity with the receiver-transmitter unit under pressure.

3.5.6.8.4 Automatic Frequency Control -

3.5.6.8.4.1 Function - The purpose of the AFC shall be to maintain the IF frequency difference between the local oscillator and the received signal for the heterodyne reception of signals.

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3.5.6.8.4.2 AFC Connection - The AFC shall connect to the crystal mixer through one RG-58/U cable terminated in suitable crystal connectors.

3.5.6.8.4.3 Manual Tuning Provisions - It shall be possible by means of an external switch located on Control, Radar Set C-2786/APS-80 to disable the automatic frequency control of the search local oscillator. In conjunction with the AFC - Manual switch there shall be a knob control which shall tune the local oscillator. There shall be a LOCAL-REMOTE switch and a MANUAL TUNE knob located on Receiver-Transmitter RT-508/APS-80 to accomplish manual tuning at the receiver-transmitter.

3.5.6.8.4.4 Frequency of Operation - The AFC shall tune and hold the frequency of the search local oscillator nominally 30 mc above the carrier-frequency.

3.5.6.8.4.5 AFC Accuracy - The AFC shall hold the frequency difference between the search local oscillator and the magnetron carrier such as to maintain an intermediate frequency amplifier, and the AFC shall hold in when the local oscillator tuning cavity is detuned to reduce crystal number 1 current to 0.2 milliampere.

3.5.6.8.5 Load Isolation - A microwave load isolator shall be provided between the magnetron and duplexer, having minimum isolation of 10 db and maximum insertion loss of 0.5 db.

3.5.6.8.6 Standing Wave Ratio - The Voltage Standing Wave Ratio of the receiver-transmitter waveguide components in the transmit direction (TR tube simulated fired) shall not exceed 1.25:1 when measured at Frequencies of 8500, 9000, 9375 and 9600 mc.

3.5.6.9 Receiver Section -

3.5.6.9.1 Function - The receiver unit shall amplify the IF signals from the mixer, detect, and by means of a cathode follower, provide video signals of the proper polarity and amplitude to the indicating system.

3.5.6.9.2 IF Amplifier Characteristics -

(1) IF shall be 30 mc $\pm$0.2 mc.

(2) IF bandwidth shall be 1.1 $\pm$0.3 mc.

(3) IF gain shall be sufficient to provide at least 1.0 volt of noise at the second detector, and shall be adjustable.

(4) The characteristic response of the receiver shall be clutter operated anti-clutter (COAC). It shall have a pedestal compression range of at least 80 db, and shall be adjustable. The receiver shall have a relay activated by a switch in the radar set control box to change receiver operation from COAC to linear operation.

(5) The IF noise figure shall not exceed 1.3 db.

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3.5.6.9.3 Recovery Time - The recovery time following an 80 db signal shall be 2 usec or less.

3.5.6.9.4 System Noise Figure - The system noise, when measured with a standard microwave noise source at the antenna connection to the receiver-transmitter shall not exceed 8.3 db when measured at 9375megacycles and shall not exceed 10.5 db when measured at frequencies of 8500, 9000, and 9600 mc with no selection of components.

3.5.6.9.5 Video Output - The receiver video output shall be of a positive polarity with respect to ground and shall be adjustable. It shall be capable of producing 2.5 volts of peak signal when terminated into 91 ohm impedance.

3.5.6.10 Transmitter Section - The transmitter shall be manually tunable and shall deliver RF in the frequency range of 8500-9600 mc/s.

3.5.6.11 Maintenance Metering Circuits - Circuits and meter shall be provided to monitor the following:

(1) AFC crystal current

(2) Radar Crystal No. 1

(3) Radar Crystal No. 2

(4) Magnetron current

(5) TR keep-alive current

3.5.7 Control, Antenna C-2785/APS-80 - Each antenna control shall be of the console, rack mounted Type 1 in conformance with Specification MIL-C-6781 and shall meet the following requirements:

3.5.7.1 Function - This control unit shall provide manual controls by means of which the operating condition of the antenna may be selected. Controlled functions shall be the tilt angle of the radar beam, antenna scan, stabilization, and heading.

3.5.7.2 Form Factor and Mounting - The dimensions and mounting of this unit shall be in accordance with Specification MIL-C-6781 and shall not exceed the following: length 5-5/8 inches, width 5-3/4 inches, and depth 7-1/4 inches, including the receptacle.

3.5.7.3 Contents - This unit shall contain the controls described in 3.5.7.5 and shall have an edge-lighted panel which shall conform to Specification MIL-P-7788. The identifying word "RADAR" shall be arranged in a vertical column at the left edge of this panel.

3.5.7.4 Weight - The weight of this unit shall not exceed 6.2 pounds.

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3.5.7.5 Controls - The following operating controls shall be provided on the front panel of this unit.

(1) "TILT" - This knob control shall permit adjustment of the mechanical tilt angle of the radar antenna with respect to its horizontal reference axis from 20° down to 10° up, by rotating the rotor(s) of the tilt synchro.

(2) "SCAN" - This multiple position knob control shall be provided to control the condition of antenna rotation. There shall be three positions - "STOP", "SECTOR", and "FULL". When placed in the "STOP" position, this control shall stop the azimuth rotation of the antenna. In the "SECTOR" position, the antenna shall scan a 45° sector. In the "FULL" position, the antenna shall normally scan 360° with PPI presentation. If the antenna is set for an adjustable sector for a nose-mounted installation, the adjustable sector shall be presented instead, with the center of the sector corresponding to the aircraft heading.

(3) "ANT HEADING" - This knob control shall permit 360° positioning or "search lighting" of the antenna by the operator when the scan switch is in the "STOP" position. It shall permit selection of any azimuth angle about which the antenna will sector scan when the "SCAN" switch, is in the "SECTOR" position.

(4) "STAB-OUT" - This two-position toggle switch shall provide for disabling or enabling the automatic tilt stabilization of the antenna reflector. When in the "STAB" position, this control shall engage the automatic tilt stabilization circuits. When in the "OUT" position, this control shall disable the automatic tilt stabilization of the antenna reflector. In both positions of the switch the antenna reflector tilt angle shall be adjustable by means of the "TILT" control. A yellow light shall be provided to indicate when the gyroscope is erecting.

3.5.8 Control, Radar Set C-2786/APS-80 - Each radar set control shall be of the console rack mounted Type I in conformance with Specification MIL-C-6781 and shall meet the following requirements:

3.5.8.1 Function - This unit shall provide controls for the operation of the radar set.

3.5.8.2 Form Factor and Mounting - The dimensions and mounting of this unit shall be in accordance with Specification MIL-C-6781 and shall not exceed the following: length 9 inches, width 5-3/4 inches and depth 6 inches.

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3.5.8.3 Contents - This unit shall contain all controls necessary for operating the transmitting and receiving functions of the radar set. The controls which pertain to the operation of the receiver shall be grouped together and those which pertain to the operation of the transmitter shall be likewise grouped together. An edge-lighted plastic panel conforming to Specification MIL-P-7788, shall cover the front surface of the control. The identifying word "RADAR" shall be arranged in a vertical column at the left edge of this panel.

3.5.8.3.1 Monitoring Provisions - Circuits and meters shall be provided to monitor the following:

(1) Magnetron Current

(2) Rectifier Current

3.5.8.4 Weight - The weight of this unit shall not exceed 5 pounds.

3.5.8.5 Controls - The following operating controls shall be provided on the front panel of this unit and shall be labeled as follows:

(1) "PWR-OFF" - This double-throw toggle switch shall turn on the equipment in a standby position. After approximately 3 minutes, the standby light shall go out.

(2) "ANT-DUMMY LOAD" - This toggle switch shall provide for normal operation of the radar antenna when in "ANT" position and shall switch the RF output of the transmitter into the dummy load when in the "DUMMY LOAD" position.

(3) "HV ON" - This momentary contact switch button applies high voltage to the magnetron. It shall operate only after the standby light goes out.

(4) "HV OFF" - This momentary contact switch shall remove power from the magnetron equipment which shall then go to standby.

(5) "AFC-MAN" - This double-throw switch shall establish automatic frequency control of the local oscillator when in the "AFC" position. When in the "MAN" position, this switch shall permit adjustment of the search local oscillator by means of the "MAN-TUNE" control.

(6) "MAN-TUNE" - This control shall provide manual frequency control by varying the value of repeller voltage of the search local oscillator over a range of 40 volts nominal, provided the "AFC-MAN" switch is in the "MAN" position.

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(7) "GAIN" - This knob control shall provide, between the limits established by adjustment of two maintenance controls, a continuously variable voltage to the cathode of the COAC control diode. This voltage shall be adjusted such that with the gain control at maximum the noise level shall be at a level which produces a 5 to 1 ratio when compared to a signal just at the saturation level. When the gain control is at a minimum, the negative bias supplied to the suppressor grids shall be adequate to provide an 80 db gain reduction.

(8) "STC" - These two knob controls shall provide for adjustment of the sensitivity time control (STC). The "RANGE" knob shall provide for an exponential bias voltage varying from 0 to 600 usec and the "DEPTH" knob shall provide for the selection of the gain reduction characteristic of the STC voltage.

(9) "LINEAR-COAC" - This double-throw switch shall provide for selection of either linear or COAC type of receiver operation.

3.5.9 Modulator-Power Supply MD-348/APS-80 - Each radar modulator shall meet the following requirements:

3.5.9.1 Function - This modulator shall accept a trigger pulse from the synchronizer, furnish the modulating pulse to the receiver-transmitter, and provide various power requirements specified herein.

3.5.9.2 Dimensions - The overall dimensions shall not exceed 22-3/4 inches height by 17-1/2 inches width by 17-1/2 inches depth.

3.5.9.3 Weight - The maximum weight shall not exceed 63 pounds.

3.5.9.4 Mounting Provisions - The unit shall be mounted on MT-2135/APS-80, a vibration isolator type shock mounting conforming to the applicable requirement of Specification MIL-C-172.

3.5.9.5 Housing and Cooling Provisions - The modulator shall be housed in a suitable can with provisions for cooling by an external fan.

3.5.9.6 Pressurization - This unit shall form a part of the pressurized system and shall be capable of withstanding an internal pressure of 30 psi, gauge.

3.5.9.7 Input Pulse - The modulator shall accept a trigger output of the synchronizer.

3.5.9.8 Modulator Pulses - Pulses for modulation of the magnetron shall meet the following requirements:

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3.5.9.8.1 Power Output - The modulator shall be capable of supplying the power required by the specific magnetron used.

3.5.9.8.2 Output Frequency and Duration - The modulator shall be capable of supplying power for the following condition:

Pulse Repetition Frequency	RF Pulse Duration
400 cycles ± 5%	2.40 ±0.12 usec

3.5.9.8.3 Shape - The shape of the modulator pulse shall be such as required for modulation of the magnetron so as to produce the output pulse characteristic specified.

3.5.9.8.3.1 Rise Time - The rise time of the modulator pulse shall be such as to produce a rate of rise of voltage at the magnetron of a minimum of 140 kv per usec and a maximum of 190 kv per usec defined as the steepest tangent above the 50% point.

3.5.9.8.3.2 Pulse Ripple - The magnetron current pulse shall have a ripple not to exceed 10%.

3.5.9.8.4 Pulse Amplitude - The modulator shall produce a pulse such that the input to the magnetron shall be a pulse from 21 to 23 kv.

3.5.9.9 Overload Protection - Overload circuits shall be provided to remove the high voltage from the modulator during excessive overload conditions. The protective circuit shall be such that on short duration loads, the circuit will reset automatically to standby.

3.5.9.10 Modulator Cooling - A 3-phase alternating current blower shall be mounted within the unit to prevent occurrence of localized hot spots and to aid, in conjunction with the external blower, in cooling the modulator.

3.5.9.11 Connectors - The modulator shall contain suitable pressurized connectors for bringing in power and trigger. A pressurized high voltage connector shall be supplied to bring out the high voltage pulse.

3.5.10 Synchronizer, Electrical SN-234/APS-80 - The electrical synchronizer shall meet the following requirements:

3.5.10.1 Function - This synchronizer shall furnish basic timing and synchronization signals to the modulators and indicators.

3.5.10.2 Form Factor and Mounting - The maximum overall dimensions shall be 7-5/8 inches height by 10-1/8 inches width by 13-7/8 inches depth. The unit shall be mounted on MT-2136/APS-80, a vibration isolator type shock mounting conforming to the applicable requirements of Specification MIL-C-172.

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3.5.10.3 Contents - This unit shall furnish eight output triggers with delay on six triggers as follows:

(1) STC 15 microseconds fixed.

(2) Strobe 24 microseconds fixed.

(3) Master indicator (2) 30 microseconds fixed.

(4) Radar modulator 15 to 32 microseconds variable.

(5) Radar indicator coincident with radar modulator.

(6) Two independent adjustments of the 15 to 32 microsecond variable delay shall be provided, relay selected from a remote position.

3.5.10.4 Weight - This unit shall not weigh more than 19.5 pounds excluding the mounting.

3.5.10.5 Electrical Connections - Electrical connections shall be provided as necessary to connect this unit to associated circuits in accordance with the requirements of this specification.

3.5.10.6 Front Panel Controls - No operating controls shall be included in this unit. The following maintenance (knob) controls shall be provided on the front panel.

(1) "Coarse Delay" - A (0-15) usec adjustable knob control.

(2) "Fine Delay" - A (0-2.0) usec adjustable knob control.

3.5.10.7 Electrical Outputs - This synchronizer shall provide the following outputs when connected to a modulator, a power source and to a control unit:

3.5.10.7.1 Modulator Trigger - In response to control signals from Data Display Group AN/ASA-16 choice of a trigger of the following characteristics shall be available:

(1) Repetition Frequencies - The synchronizer shall furnish trigger pulses at a repetition frequency of 400 cps ±5%.

(2) Amplitude - The peak amplitude shall be not less than 25 volts.

(3) Shape - The duration at 50% of maximum amplitude shall not be more than 2 usec and the rate of rise such that the 90% of maximum amplitude point shall be reached in not more than 0.5 usec.

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(4) Impedance - These characteristics shall be met when this output is loaded with 1800 to 2200 ohms resistance and up to 400 micromicrofarads shunt capacitance at the end of 60 feet of AN type RG-62/U cable.

3.5.10.7.2 Master Ind. Out - There shall be two of these outputs. Each output shall be not less than 25 volts.

3.5.10.7.3 STC Out - This output shall be a 25-volt trigger minimum.

3.5.10.7.4 Strobe Out - This output shall be a 25-volt trigger minimum.

3.5.10.7.5 IFF Out - This output shall be a 25-volt trigger minimum.

3.5.10.7.6 Slave Ind. Out - This output shall be a 25-volt trigger minimum.

3.5.10.7.7 Sync Out - This output shall be a 25-volt trigger minimum.

3.5.11 Gyroscope, Displacement CN-549/APS-80 or CN-611/APS-80A - Each gyroscope shall meet the following requirements:

3.5.11.1 Function - This unit shall provide output voltages for tilt stabilization of the reflector of stabilized antenna assemblies. It shall be an approved electrical and mechanical equivalent to commercial type Bendix 14128-4-A.

3.5.11.2 Dimensions - The outline dimensions shall not exceed 5-7/8 inches height by 6-1/2 inches width by 9-1/8 inches depth.

3.5.11.3 Weight - The weight shall not exceed 9.0 pounds.

3.5.11.4 Mounting Provisions - This unit shall be capable of mounting to a flat surface on the antenna or elsewhere on the aircraft.

3.5.11.5 Contents - The gyroscope unit shall be a two axis gyro transmitter consisting of a dust sealed case containing a vertical gyroscope with erection switches and associated torque motors.

3.5.11.6 Characteristics - This unit shall have the following characteristics:

3.5.11.6.1 Speed - The gyroscope shall reach its rated speed of at least 22,000 rpm within 5 minutes, after being energized in an ambient temperature of 25°C.

3.5.11.6.2 Erection Rates - There shall be two erection rates, selected automatically inside the unit, which depend on the displacement from true vertical. Correction rates shall be approximately 20° per minute for fast erection and approximately 2° per minute for normal erection.

3.5.11.6.3 Roll Brake - A roll brake shall be incorporated to prevent tumbling after power has been removed.

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3.5.11.6.4 Leveling - One or more surfaces shall be included which will provide areas on which a spirit level can be placed in order to establish a plane perpendicular to gyroscope vertical. These areas shall be labeled "REF LEVEL".

3.5.11.6.5 Switches - The erection switches of the gyro shall be energized when the aircraft is in straight and level flight. A rate switch shall be incorporated in the circuit to cut off the erection voltage when the rate of turn of the aircraft is within the limits specified in Paragraph 3.5.16.1.

3.5.12 Fan, Tubeaxial HD-359/APS-80 - Each fan shall meet the following requirements:

3.5.12.1 Function - Four 3-phase alternating current fans shall be provided to supply cooling air to each Modulator-Power Supply MD-348/APS-80 and Receiver-Transmitter, Radar RT-508/APS-80.

3.5.12.2 Dimensions - The maximum overall dimensions including the funnel shaped output duct shall not exceed 7-5/8 inches length by 5-5/16 inches width by 6-11/16 inches depth.

3.5.12.3 Weight - The weight of each fan shall not exceed 3.5 pounds.

3.5.12.4 Mounting Provisions - The unit shall be externally mounted by a mounting bracket which shall be an integral part of the fan.

3.5.13 Power Supply PP-2192/APA-125A - Each power supply shall meet the following requirements:

3.5.13.1 Function - The power supply shall contain transformers, rectifiers, filters and regulators necessary to deliver the power required by the receiver-transmitter, synchronizer and synchro signal amplifier. This unit shall not contain servo amplifiers.

3.5.13.2 Dimensions and Mounting - The overall dimensions shall not exceed 13 inches width by 10-3/4 inches height by 19-1/4 inches depth. The unit shall be mounted on MT-1824/APA-125.

3.5.13.3 Weight - The power supply weight shall not exceed 40 pounds, exclusive of mounting.

3.5.13.4 Outputs - The power supply shall be capable of delivering the following:

(1) +150 V dc, regulated +150 mv at 200 ma. nominal.

(2) -150 V dc, regulated ±150 mv at 400 ma. nominal.

(3) +250 V dc, regulated ±250 mv at 300 ma. nominal.

(4) +300 V dc, regulated ±3 V at 250 ma. nominal.

(5) -300 V dc, regulated ±300 mv at 20 ma. nominal.

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3.5.14 Switch, Waveguide SA-512/U - Each waveguide switch shall meet the following requirements:

3.5.14.1 Function - The waveguide switch shall connect the transmitter RF output to Dummy Load, Electrical DA-148/U when the switch on Control, Radar Set C-2786/APS-80 is placed in the "Dummy Load" position.

3.5.14.2 Dimensions - The overall dimensions shall not exceed 4-7/8 inches height by 4-5/16 inches width by 6-1/16 inches depth.

3.5.14.3 Weight - The weight shall not exceed 5.0 pounds.

3.5.14.4 Electrical Connections - The following electrical connections shall be furnished:

(1) Waveguide to coupler.

(2) Waveguide to antenna.

(3) Waveguide to Dummy Load.

(4) Control Wire - +28 volts.

3.5.14.5 VSWR - The VSWR shall not exceed 1.12 over the frequency range of 9400 to 9600 mc.

3.5.14.6 Isolation - With the radar RF output switched to the dummy load, the isolation between the RF input and the antenna connection shall be at least 50 db.

3.5.15 Dummy Load, Electrical DA-148/U - Each electrical dummy load shall meet the following requirements:

3.5.15.1 Function - The dummy load shall be capable of providing a termination for the transmitter when the RF channel is switched to the dummy load position.

3.5.15.2 Dimensions - The overall dimensions shall not exceed 3-1/4 inches height by 2-5/8 inches width by 6-1/2 inches depth.

3.5.15.3 Weight - The weight shall not exceed 1.5 pounds.

3.5.15.4 Power - The dummy load shall be capable of absorbing 275 kw peak power and 250 watts average power.

3.5.15.5 VSWR - The VSWR shall not exceed 1.12 over the frequency range of 8500 to 9600 mc/s.

3.5.15.6 Electrical Connections - The dummy load shall be connected to the transmitter side of the RF output by means of the dummy load termination of the Switch, Waveguide SA-512/U.

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3.5.16 Gyroscope, Rate CN-550/APS-80 or CN-612/APS-80A - Each rate gyroscope shall meet the following requirements:

3.5.16.1 Function - This unit shall cut off the displacement gyroscope erection voltage when the rate of turn of the aircraft is between 10° and 25° per min. for CN-550/APS-80 or 40° and 55° per min. for CN-612/APS-80A.

3.5.16.2 Dimensions - The overall dimensions shall not exceed 4-3/16 inches height by 3-1/4 inches width by 5-7/16 inches depth.

3.5.16.3 Weight - The weight shall not exceed 2.5 pounds.

3.5.16.4 Mounting Provisions - This unit shall be mounted on the Antenna AS-1000/APS-80.

3.5.17 Interconnection Box J-1192/APS-80 - Each interconnection box shall meet the following requirements:

3.5.17.1 Function - This unit shall provide wiring and connectors for necessary interconnection of cables between Amplifier, Synchro Signal AM-2719/APS-80 and the following units: Antenna AS-1000/APS-80, Modulator MD-348/APS-80, Receiver-Transmitter RT-508/APS-80 and Fans, Tubeaxial HD-359/APS-80.

3.5.17.2 Dimensions - The overall dimensions shall not exceed 12-1/4 inches in height by 10 inches width by 4 inches depth.

3.5.17.3 Weight - The weight of this unit shall not exceed 8 pounds.

3.5.17.4 Mounting Provisions - This unit shall mount on a flat surface on 5 by 8 inch centers.

3.5.18 Interconnection Box J-1035/APS-80 - The interconnection box shall meet the following requirements:

3.5.18.1 Function - This unit shall provide wiring and connectors for necessary interconnecting of cables for AN/APS-80 Radar Set, AN/APA-125 Indicator Group, and the Interconnecting Box of the AN/ASA-25.

3.5.18.2 Dimensions - The overall dimensions shall not exceed 25 inches in height by 9-1/2 inches width by 3 inches depth.

3.5.18.3 Weight - The weight of this unit shall not exceed 15 pounds.

3.5.18.4 Mounting Provisions - This unit shall mount on a flat surface without shock mounts.

3.5.19 Transformer, Variable Power TF-284/APS-80 - Each variable power transformer shall meet the following requirements:

3.5.19.1 Function - This 3-phase unit shall provide a manual control for setting correct magnetron current and shall contain a running time totalizer for magnetron operating time.

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3.5.19.2 Dimensions - The overall dimensions shall not exceed 7-3/8 inches in height by 9-1/8 inches width by 12-1/8 inches depth.

3.5.19.3 Weight - The weight of this unit shall not exceed 20 pounds.

3.5.19.4 Mounting Provisions - This unit mounts on four self-contained vibration isolators.

4. QUALITY ASSURANCE PROVISIONS

4.1 General - Items covered by this specification shall be subjected to the following tests to determine compliance with all applicable requirements:

(1) Preproduction Tests

(2) Production Sample Tests

(3) Acceptance Tests

(4) Life Tests

4.2 Preproduction Tests - Preproduction tests shall be made on one or more equipments representative of the production equipments to be supplied under the contract. Preproduction tests shall consist of the following:

(1) Contractor's Demonstration Tests

(2) Service Approval Tests

4.2.1 Contractor's Demonstration Tests - Contractor's demonstration tests shall be accomplished under the responsibility of the contractor and shall be conducted in accordance with the approved test procedure of 4.6. Data obtained by the contractor in conducting these tests shall be submitted to the procuring activity for review and approval prior to shipping the equipment to the specified destination for service approval tests. The government inspector and the procuring activity shall be advised when tests are to be conducted so that a representative may be designed to witness or supervise the tests when so desired. Contractors not having adequate facilities to conduct all required tests shall obtain the service of a commercial testing laboratory satisfactory to the procuring activity.

4.2.2 Service Approval Tests - At the completion of the contractor's demonstration tests and when requested by the procuring activity, the equipment shall be delivered to a specified government laboratory for additional testing. This additional testing may consist of duplicating tests previously conducted and such other tests as are deemed necessary to determine compliance with all applicable design and performance requirements.

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4.2.2.1 Accessory Material - In addition to the complete equipment submitted for service approval tests, the contractor shall also submit the accessory material and design and test data specified in Specification MIL-E-5400. This information shall indicate the physical and electrical characteristics of the equipment and establish the equipment's compliance with applicable requirements.

4.2.3 Scope of Tests - Preproduction tests shall include all tests deemed necessary to determine that the equipment meets all the requirements of this specification and the contract. Preproduction tests shall include environmental tests in accordance with Specification MIL-T-5422 and interference tests in accordance with Specification MIL-I-6181.

4.2.4 Preproduction Approval - Approval of the preproduction model shall be by the procuring activity upon satisfactory completion of all tests. No production equipments shall be delivered prior to the approval of the preproduction model. Prefabrication of production equipment prior to the approval of the preproduction model is at the contractor's own risk. The approved preproduction model will be returned to the contractor for his use in the fabrication and testing of equipment to be submitted for acceptance. The preproduction model shall not be considered as one of the equipments under the contract; however, it may be reworked by the contractor and submitted for acceptance as a production equipment.

4.3 Production Sample Tests - When requested by the procuring activity, one equipment shall be selected from the first 10 production equipments submitted for acceptance and sent to a designated government laboratory for tests. The equipment shall be selected by the government inspector after the equipment has successfully passed all individual tests.

4.3.1 Scope of Tests - This equipment may be subjected to any and all tests the procuring activity deems necessary to assure that the production equipment is equivalent to the previously approved preproduction sample in design, construction and performance and that it meets all applicable requirements.

4.3.2 Production Sample Approval - Approval of the production sample shall be by the procuring activity upon satisfactory completion of all tests. Any design, material or performance defect made evident during this test shall be corrected by the contractor to the satisfaction of the procuring activity. Upon failure of the production sample to pass any of the tests, delivery of equipment under the contract shall cease until proper corrective action is approved and accomplished. Corrective action shall also be accomplished on equipment previously accepted when requested by the procuring activity.

4.3.3 Reconditioning of Production Test Sample - On completion of the production sample test the equipment shall be reworked by the contractor by replacing all wear or damaged items. After reworking the contractor shall resubmit the equipment for acceptance.

4.4 Acceptance Tests - The contractor shall furnish all samples and shall be responsible for accomplishing the acceptance tests. All inspection and testing shall be under the supervision of the government inspector. Contractors

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not having testing facilities satisfactory to the procuring activity shall engage the service of a commercial testing laboratory acceptable to the procuring activity. The contractor shall furnish test reports showing quantitative results for all acceptance tests. Such reports shall be signed by an authorized representative of the contractor or laboratory, as applicable. Acceptance or approval of material during the course of manufacture shall not be construed as a guarantee of the acceptance of the finished product. Acceptance tests shall consist of the following:

(1) Individual Tests

(2) Sampling Tests

(3) Special Tests

4.4.1 Individual Tests - Each equipment submitted for acceptance shall be subjected to the individual tests. These tests shall be adequate to determine compliance with the requirements of material, workmanship, operational adequacy and reliability. As a minimum, each equipment accepted shall have passed the following tests:

(1) Examination of Product

(2) Operational Test

(3) Manufacturing Reliability Test

4.4.1.1 Examination of Product - Each equipment shall be examined carefully to determine that the material and workmanship requirements have been met.

4.4.1.2 Operational Test - Each equipment shall be operated long enough to permit the equipment temperature to stabilize and to check sufficient characteristics and record adequate data to assure satisfactory equipment operation.

4.4.1.3 Manufacturing Reliability Test - Each equipment shall be operated under the conditions specified herein for a period of 6 hours without failure. A failure shall be defined as anything which causes malfunctioning of the equipment. Only those adjustments will be permitted which can be made by using such controls and adjustments that are accessible to the operator during the normal use of the equipment.

Temperature	Ambient room
Humidity	Ambient room
Vibration (10 minutes)	Any selected frequency within the range of 20 to 30 cps (excluding resonant points) and a minimum amplitude of ± 3 g's

The equipment shall be vibrated (without vibration isolators) for a period of 10 minutes prior to the beginning of the 6-hour period of operation. Where feasible,

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the equipment shall be operated during this vibration period for the purpose of detecting flaws and imperfect workmanship. Operation within the specified limits of satisfactory performance is not necessarily required during the vibration period. The direction of vibration should be vertical to the normal mounting plane for 5 minutes and lateral to that plane for 5 minutes. Where it is not feasible to vibrate the equipment in 2 directions the vertical direction shall be used. During the 6-hour period of operation following the 10-minute vibration period, the equipment shall be mechanically cycled periodically through its various phases of operation. Should a failure occur, it should be repaired and the test started over, except that the 10-minute vibration period need not be repeated when it is certain the failure was not a result of the vibration. Should repetitive failures occur, corrective action shall be taken to eliminate this defect from future equipment. A record shall be kept of all failures. The 6-hour period specified above may be composed of two 3-hour periods to conform with standard working hours.

4.4.2 Sampling Tests - Sampling tests shall be conducted on equipments which have successfully passed the individual tests. The following equipments shall be selected by the Government Inspector for sampling tests:

Quantity of Equipments Offered for Acceptance	Quantity to be Tested
First 10	1
Next 50	1
Next 75	1
Next 100	1
	1 for each additional 200 or fraction thereof

4.4.2.1 Scope of Tests - As a minimum, each equipment selected for sampling tests shall be subjected to the following tests:

(1) Complete operational test at ambient room conditions, making all necessary measurements to assure that all applicable specification requirements have been met.

(2) Operational test at certain environmental conditions. The conditions may vary for each equipment tested and should be based on results of the preproduction and special tests.

(3) Manufacturing reliability test specified in paragraph 4.4.1.3 except that the test duration shall be 120 hours with no restriction on the number of failures. However, each failure shall be analyzed as to cause and remedial action necessary to reduce the possibility of its recurrence in future equipment.

4.4.3 Special Tests - Special tests shall be conducted on a quantity

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of equipments for the purpose of checking the effect of any design or material change on the performance of the equipment and to assure adequate quality control. The equipment selected for special tests may be selected from equipments previously subjected to the sampling tests.

4.4.3.1 Special Tests Schedule - Selection of equipments for special tests shall be made as follows:

(1) On an early production equipment.

(2) On an equipment midway of the production run.

(3) On an early equipment after an engineering change.

(4) Whenever failure reports or other information indicate additional equipments should be tested. (This will be determined by the procuring activity).

Exceptions from the above may be necessary when the number of equipments procured under contract or other factors affecting performance may require a variation to assure satisfactory control.

4.4.3.2 Scope of Tests - Special tests shall consist of such tests as are approved by the procuring activity. Test procedures previously approved for the preproduction and sampling tests shall be used where applicable. When not applicable, the contractor shall prepare a test procedure and submit it to the procuring activity for approval prior to conducting the tests. As a minimum, the following tests shall be made, except that for those equipments selected as a result of design changes and failure reports, only those tests required to check the characteristics in question need be conducted.

(1) Temperature - Altitude Tests

(2) Vibration Test (60 minutes cycling in each of 3 planes)

(3) Humidity Tests

4.4.4 Equipment Failure - Should a failure occur during either the sampling or special tests, the following action shall be taken:

(1) Determine the cause of failure.

(2) Determine if the failure is an isolated case or design defect.

(3) Submit to the procuring activity for approval, proposed corrective action intended to reduce the possibility of the same failure(s) occurring in future tests.

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(4) Where practical, include a test in the individual test to check all equipment for this requirement until reasonable assurance is obtained that the defect has been satisfactorily corrected.

4.5 Life Test - A 300-hour life test shall be conducted at the contractor's plant on an early equipment representative of the current production. The life test shall be performed under the conditions specified in 4.5.1. The life test sample shall be selected by the government inspector in accordance with the following: When a production sample equipment is submitted to the procuring activity for test then the first equipment specified below need not be selected or subjected to this test. (Equipments which have successfully passed the sampling tests or special tests may be selected for life tests).

Quantity of Equipments Offered for Acceptance	Quantity to be Tested
First 25	1
Next 175	1
Next 300	1
	1 for each additional 500 or fraction thereof

4.5.1 Test Conditions - The life test shall be conducted under the following simulated service conditions.

Temperature	Normal room
Altitude	Normal ground (0 - 5000 ft.)
Humidity	Room ambient
A.C. voltage	115 ± 5 volts (at lowest applicable frequency)
D.C. voltage	27.0 ± 2.0 volts

4.5.2 Test Periods - The test may be run continuously or intermittently. Any period of operation shall be of sufficient duration to permit the equipment temperature to stabilize. Periodically, the equipment shall be turned on and off several times and put through its various phases of operation.

4.5.3 Performance Check - At approximately 8 hour intervals during the test, a limited performance check shall be made. The performance check proposed by the contractor shall be subject to approval by the procuring activity.

4.5.4 Test Data - The contractor shall keep a daily record of the performance of the equipment, making particular note of any deficiencies or failures. In the event of part failures, the defective part shall be replaced and the operation resumed for the balance of the test period. A record shall be kept of all failures throughout the test, including all tube failures. This record shall indicate the following:

(1) Part type number

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(2) The circuit reference symbol number

(3) The part function

(4) Name of the manufacturer

(5) Nature of the failure

(6) The number of hours which the part operated prior to failure

4.5.4.1 Failure Report - In the event of a failure, the government inspector shall be notified immediately. A report shall be submitted to the procuring activity upon completion of the test. In this report, the contractor shall propose suitable and adequate design or material corrections for all failures which occurred. The procuring activity will review such proposals and determine whether they are acceptable.

4.5.5 Reconditioning of Life Test Samples - An equipment which has been subjected to the life test shall be reconditioned as follows:

(1) On completion of the life test, the equipment shall be reworked by the contractor by replacing all tubes and "wear" items. The "wear" items shall be determined by agreement between the contractor and the procuring activity.

(2) After reworking, the contractor may resubmit the equipment for acceptance.

4.6 Test Procedures - The procedures used for conducting preproduction tests, acceptance tests and life tests shall be prepared by the contractor and submitted to the procuring activity for review and approval. The right is reserved by the procuring activity or the government inspector to modify the tests or require any additional tests deemed necessary to determine compliance with the requirements of this specification or the contract. Specification MIL-T-18303 shall be used as a guide for preparation of test procedures. When test procedures are available from previous contracts such procedures may be used when approved by the procuring activity.

4.7 Presubmission Testing - No item, part or complete equipment shall be submitted by the contractor until it has been previously tested and inspected by the contractor and found to comply, to the best of his knowledge and belief, with all applicable requirements.

4.8 Rejection and Retest - Equipment which has been rejected may be reworked or have parts replaced to correct the defects and resubmitted for acceptance. Before resubmitting, full particulars concerning previous rejection and the action taken to correct the defects found in the original shall be furnished the government inspector. Units rejected after retest shall not be resubmitted without the specific approval of the procuring activity.

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4.9 Equipment Changes - After approval of the preproduction model, no changes which affect weight, installation interchangeability and interchangeability of maintenance parts or assemblies shall be incorporated in the equipment unless approved by the procuring activity. To obtain procuring activity approval, the contractor shall submit an Engineering Change Proposal (ECP) giving the necessary information concerning the change. This shall apply for either a procuring activity's recommended change or a contractor's proposed change. When required, the contractor shall also prepare a Material Change (MC) and a Material Bulletin (MB) to cover the approved change. Engineering Change Proposals, Material Changes and Material Bulletins shall be prepared in accordance with SAR-300.

5. PREPARATION FOR DELIVERY

5.1 General - All major units and parts of the equipment shall be preserved, packaged, packed and marked for the level of shipment specified in the contract or order in accordance with Specification MIL-P-17555.

6. NOTES

6.1 Test Values - Normal and limiting values of performance data shall be determined at input voltages of 27.5 ± 0.5 V DC and 115 ± 1.0 V AC. These data are to be used in testing the equipment at installation points for compliance with minimum acceptable standard of performance.

6.2 Precedence of Documents - When the requirements of the contract, this specification, or applicable subsidiary specifications are in conflict, the following precedence shall apply:

(1) Contract - The contract shall have precedence over any specification.

(2) This Specification - This specification shall have precedence over all applicable subsidiary specifications. Any deviation from this specification, or from subsidiary specifications where applicable, shall be specifically approved in writing by the procuring activity.

(3) Referenced Specifications - Any referenced specification shall have precedence over all applicable subsidiary specifications referenced therein. All referenced specifications shall apply to the extent specified.

6.3 The parentheses (*), when used in the type designation, shall be replaced by either a number or letter furnished by the procuring activity upon application by the contractor for assignment of nomenclature in accordance with 3.2.15. The complete type number shall be used on nameplates, shipping records and instruction books, as applicable.

6.4 Performance Objectives - Minimum size and weight, simplicity

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of operation, ease of maintenance, and an improvement in the performance and reliability of the specific functions beyond the requirements of this specification are objectives which shall be considered in the production of this equipment. Where it appears a substantial reduction in size and weight or improvement in simplicity of design, performance, ease of maintenance or reliability will result from the use of materials, parts and processes other than those specified in Specification MIL-E-5400, it is desired their use be investigated. When investigation shows advantages can be realized, a request for approval shall be submitted to the procuring activity for consideration. Each request shall be accompanied by complete supporting information.

6.5 The primary input power requirements defined in 3.2.13 changes the requirements as defined in Specification MIL-E-7894 to requirements as defined in MIL-STD-704. Properly assigned category will require no design changes of the equipment.

NOTICE - When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished or in any way supplied the said drawings, specifications or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use or sell any patented invention that may in any way be related thereto.

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